

Derek Nagel

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PROFILE

Mechanical Engineering student with a concentration in Aerospace. Completed a summer 2026 Mechanical Engineering internship at Astronics Connectivity Systems & Certification — **RTCA DO-160G environmental qualification testing**, structural FEA, and manufacturing automation inside AS9100 / ITAR-controlled programs — and earned a return offer. Hands-on builder across mechanical, electrical, and software projects; licensed *pilot*, single and multi-engine.

EDUCATION

George Mason University

B.S. Mechanical Engineering with Concentration in Aerospace
Minor — Aviation Flight Training and Management

Fairfax, VA

Expected Graduation — May 2028

Completed

Relevant Coursework: Statics, Dynamics, Thermodynamics, Solid Mechanics, Design of Mechanical Elements, Circuit Theory

GPA: 3.65 / 4.0 | *Honors:* Presidential Academic Scholarship

TECHNICAL SKILLS

SolidWorks, ANSYS, MATLAB, Autodesk Inventor, Python, FEA & Modal Analysis, Engineering Drawings & GD&T, Tolerance Analysis, Root Cause & Corrective Action, First Article Inspection, Statistical Process Control, HALT/HASS Accelerated-Life Testing, DO-160G Qualification, Vibration & Temp-Humidity Testing, CAN Bus Reverse-Engineering, Excel Cost Modeling & Data Visualization, AI-Assisted Engineering Automation (Claude), EV & HV Electrical Systems, Private Pilot (Single & Multi-Engine)

PROFESSIONAL EXPERIENCE

Astronics Connectivity Systems & Certification — Mechanical Engineering Intern

May 2026 – Aug 2026

Waukegan, IL

Return offer received

- Returned a Qualmark Typhoon 3.0 **HALT/HASS environmental test chamber** to service after an extended outage: root-caused the 480 V power-up fault, commissioned thermal (LN2) and 6-DOF vibration operation, and authored the procedures that restored in-house **accelerated-life testing** — thermal step-stress, rapid thermal cycling, vibration step-stress.
- Rebuilt an inaccurate vendor surface model into a **SolidWorks** structural model of a 4-MCU ARINC 600 tray and ran **FEA modal analysis** (SolidWorks, ANSYS) against a 50 Hz first-natural-frequency airframe requirement, then validated it with in-house **vibration testing** benchmarked against an accredited external lab.
- Supported **RTCA DO-160G** qualification campaigns spanning Sec. 8 **vibration** and multi-day Sec. 6 **temperature-humidity** testing at an accredited external lab, correlating lab results against in-house chamber data.
- Built two **capital-equipment business cases** presented to engineering leadership — a **robotic riveting cell** (136 flush rivets per plate) and an in-house placard cell covering 15,000+ annual units across 400 part numbers — with time studies, throughput modeling, equipment selection, and **CAPEX/payback modeling in Excel**.
- Supported **aerospace LRU mechanical design** for connectivity hardware flying on Airbus and Boeing platforms — **AS9100** processes, **first article inspection**, **statistical process control**, FAA TSO/PMA certification, **ITAR**-controlled programs.

Tesla — Tesla Advisor (Seasonal)

May 2025 – Sep 2025

Schaumburg, IL

- Supported **end-to-end electric vehicle delivery operations** — late-stage staging, quality checks, and final inspection — and studied **battery production**, electric motor components, and assembly-line quality control at Tesla's **Elgin manufacturing facility**.

Independent Study in Electrical, Mechanical, and Software Design

Aug 2022 – Present

- Tesla Model S Re-build* — Disassembled and rebuilt a heavily corroded battery pack and drivetrain on a 260,000-mile 2014 Tesla Model S P85; refurbished the pack with a custom corrosion treatment and returned the car to the road.
- Plane Pointer* — Designed and built a 2-axis automated tracking system that physically points at live aircraft from ADS-B data, displaying flight ID, position, altitude, and speed.
- Pre-Autopilot Driver-Assistance Retrofit* — Technical lead of a three-person team retrofitting **openpilot** onto a pre-Autopilot Tesla Model S: reverse-engineered the power-steering rack's **CAN bus protocol** by bench and in-vehicle sniffing, built a Python toolkit for closed-loop steering commands, and ported the vehicle interface to current comma hardware.
- Smells Phishy* — Sole owner and developer of an AI scam-detection product: native **iOS app** (Swift) over a Next.js backend, carried from architecture through App Store release engineering.

EXTRACURRICULAR

Division I Athlete, Men's Volleyball and President, Athletes in Action — George Mason University. Compete in a D1 program alongside a full engineering course load; lead a student-athlete chapter of gatherings, service, and outreach.